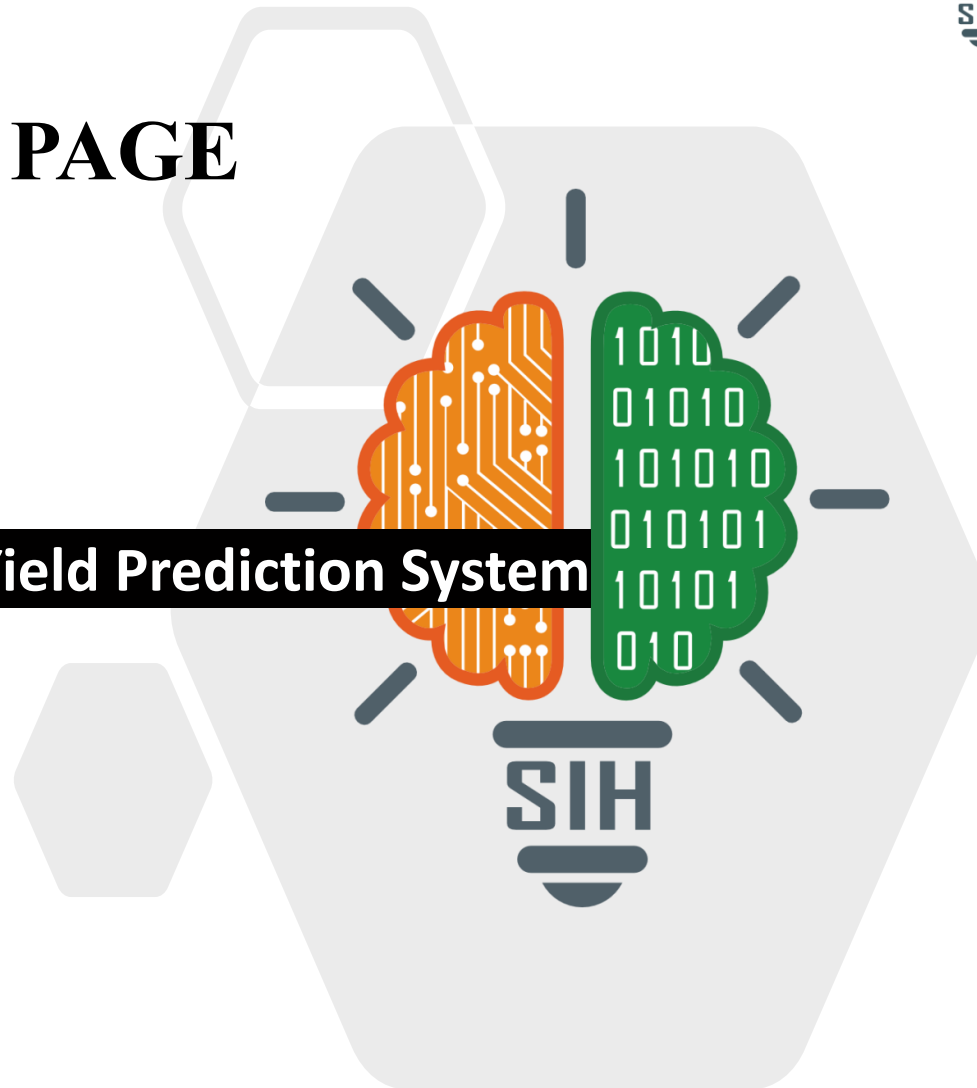


TITLE PAGE

- Problem Statement ID – 9
- Problem Statement Title-

AI-Based Crop Disease Detection and Yield Prediction System

- Theme- Agri-Tech
- PS Category- Software
- Team ID- 2
- Team Name EventLoop



Idea / Solution

Implementation of an **AI-powered smart farming platform** for intelligent crop monitoring and decision-making.

- ◆ **AI-based crop disease detection** using crop images
- ❖ **ML-based yield prediction** using farm & crop data
- ❖ **Gemini-powered AI assistant** for agricultural guidance
- ❖ **Real-time weather & agriculture insights**
- ❖ **Hindi-based recommendations** for better accessibility

Problem Resolution:

- ◆ **Early disease detection** reduces crop loss and improves timely intervention.
- ❖ **Yield prediction** helps farmers plan resources and harvest decisions.
- ❖ **Unified platform** eliminates the need for multiple disconnected tools.

Unique Value Propositions (UVP):

- ◆ **AI-powered Disease → Diagnosis → Recommendation**
- ❖ **Farm-specific intelligent insights** using crop, soil & location data
- ◆ **Multimodal AI** combining Computer Vision, ML & Gen AI
- ❖ **Actionable recommendations in Hindi** for farmer accessibility

AI & ML DEVELOPMENT:



Disease Detection: TensorFlow/Keras — Image-based crop disease classification.



Yield Prediction: Scikit-learn — Crop yield prediction using farm & disease data.



Generative AI: Gemini API — Disease insights, recommendations & AI assistance.

WEB APPLICATION DEVELOPMENT:



Backend: Flask — REST APIs, authentication & AI integration.

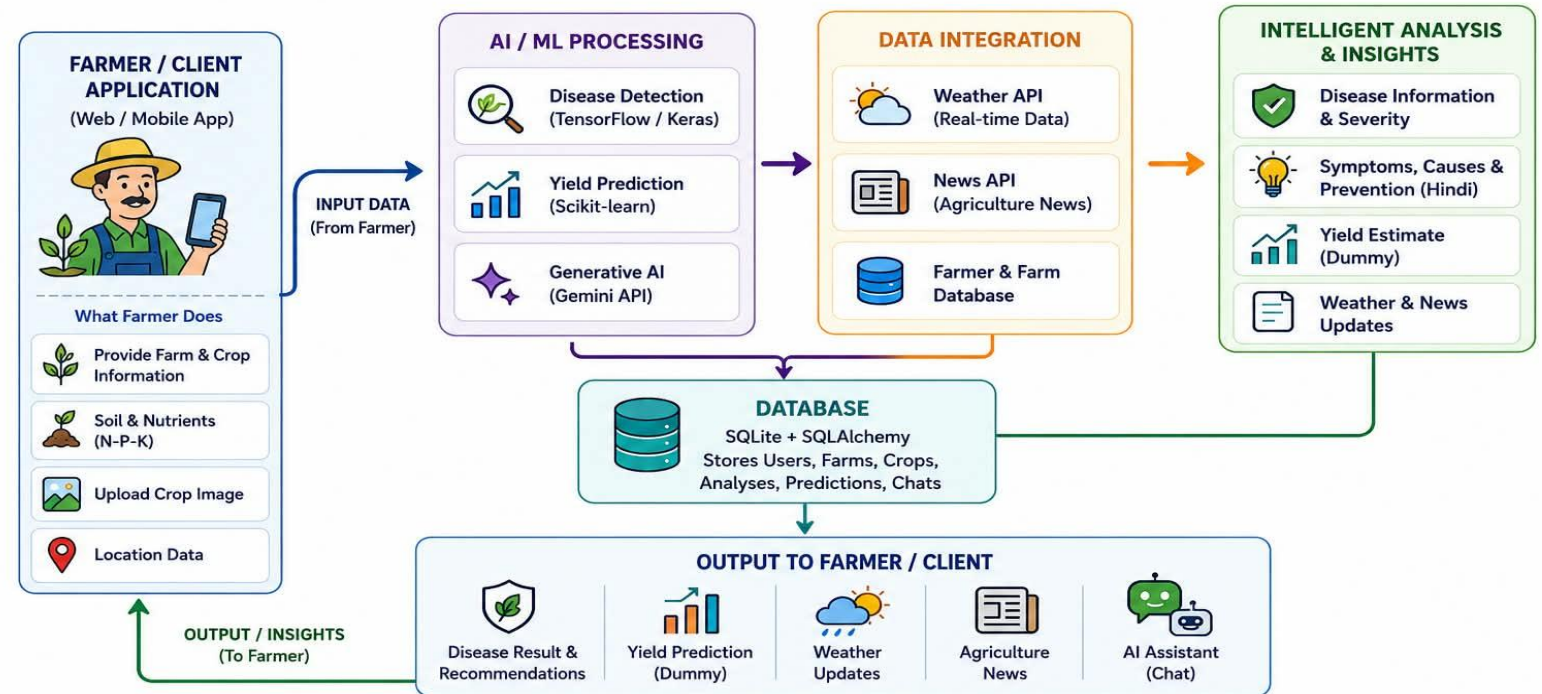


Frontend: HTML, CSS & JavaScript — Mobile Phone Responsive farmer interface. **** (PWA)**



Database: SQLite + SQLAlchemy — Farm, crop & prediction data management.

PROCESS FLOW ARCHITECTURE



Product Status

~85% Product Completed — Core AI, Farm Management, Weather, News & Image Analysis modules are implemented and functional. **Yield Prediction is currently in Demo/Dummy mode**; testing and final model integration are in progress. **** Progressive Web App**

TECHNICAL FEASIBILITY



- Built using Flask, TensorFlow, Scikit-learn & Gemini API
- Modular architecture enables easy integration and scaling
- Lightweight web platform accessible on smartphones

ECONOMIC VIABILITY



- Uses open-source technologies to minimize development cost
- Cloud/API-based architecture supports scalable deployment
- Can reduce crop losses through **early detection & better decisions**

OPERATIONAL FEASIBILITY



- Simple image-based disease detection for farmers
- Hindi AI recommendations improve accessibility
- Centralized farm, crop, weather & disease information

SCALABILITY & FUTURE SCOPE



- Extend to **more crops, diseases & regional languages**
- Integrate IoT sensors, satellite data & advanced yield models
- Progressive Web App enables wider rural accessibility

FARMER IMPACT



- **Early disease detection** → reduces potential crop losses
- **Better yield planning** through AI-based predictions
- **Hindi guidance** makes technology more accessible

ECONOMIC BENEFITS



- *Reduces unnecessary crop treatment costs*
- *Improves resource utilization and farm productivity*
- *Supports data-driven farming decisions*

AGRICULTURAL IMPACT



- Promotes **smart & sustainable farming**
- Enables timely weather and crop-related decisions
- Helps farmers move from reactive to **preventive crop management**

LONG-TERM IMPACT



- Scalable across **crops, regions and languages**
- Foundation for IoT, satellite & precision-agriculture integration

1. **Mohanty, S. P., Hughes, D. P. & Salathé, M. (2016)**
Using Deep Learning for Image-Based Plant Disease Detection
→ Foundation for AI-based crop disease detection using images.
2. **van Klompenburg, T., Kassahun, A. & Catal, C. (2020)**
Crop Yield Prediction Using Machine Learning: A Systematic Literature Review
→ Supports ML-based crop yield prediction using agricultural factors.
3. **Upadhyay, A. et al. (2025)**
Deep Learning and Computer Vision in Plant Disease Detection
→ Recent research on computer vision and deep learning in precision agriculture.
4. **Jabed, M. A. & Murad, M. A. A. (2024)**
Crop Yield Prediction in Agriculture: A Comprehensive Review of ML & DL Approaches
→ Supports data-driven yield prediction and sustainable agriculture.